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Track: AI & Data Science - Microsoft Machine Learning Engineer

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Project Name

"AI-Powered Predictive Maintenance for Industrial Equipment"

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Requirements Gathering

Introduction:

- **Purpose**
  - The purpose of this document is to define, document, and communicate the requirements of the Explainable AI-Driven Predictive Maintenance System. It serves as a reference for project stakeholders, developers, testers, and evaluators by describing the system objectives, scope, user requirements, functional and non-functional requirements, interfaces, constraints, assumptions, and acceptance criteria. This document ensures a shared understanding of the project requirements and provides a foundation for system design, implementation, testing, deployment, and validation of the predictive maintenance system, including its simulated real-time monitoring capabilities.

- **Scope**
  - This document covers the requirements for the Explainable AI-Driven Predictive Maintenance System, including stakeholder needs, user stories, use cases, functional and non-functional requirements, system interfaces, constraints, assumptions, acceptance criteria, and quality requirements. It also defines the requirements for simulated real-time data streaming, Remaining Useful Life (RUL) prediction, explainable AI analysis, backend services, and interactive dashboard visualization. This document serves as the primary reference for project planning, development, testing, deployment, and evaluation throughout the project lifecycle.

- ## Definitions and Acronyms

Term	Meaning	Description
AI	Artificial Intelligence	The simulation of human intelligence by computer systems
XAI	Explainable Artificial Intelligence	Techniques that make machine learning predictions understandable to users
ML	Machine Learning	Algorithms that learn patterns from data and make predictions
RUL	Remaining Useful Life	The estimated number of operational cycles remaining before equipment failure
CMAPSS	Commercial Modular Aero-Propulsion System Simulation	NASA's engine degradation simulation dataset used for predictive maintenance research
SHAP	SHapley Additive exPlanations	A model explainability technique that measures feature contributions to predictions
API	Application Programming Interface	A software interface used for communication between system components
EDA	Exploratory Data Analysis	The process of analyzing and visualizing data to understand patterns and relationships
KPI	Key Performance Indicator	A measurable metric used to evaluate project success
PHM	Prognostics and Health Management	Techniques used to predict equipment health and future failures
REST	Representational State Transfer	An architectural style used for designing web APIs
JSON	JavaScript Object Notation	A lightweight data format used for API communication
RMSE	Root Mean Square Error	A regression metric used to measure prediction error
MAE	Mean Absolute Error	The average absolute difference between predicted and actual values
R^2	Coefficient of Determination	A metric that indicates how well a model explains the variance in the target variable

Stakeholder Analysis:

- **Maintenance Engineers**
  - Role:
    - Maintenance engineers are the primary users of the predictive maintenance system. They are responsible for monitoring equipment health, scheduling maintenance activities, and minimizing unexpected failures.
  - Responsibilities:
    - Monitor equipment health status.
    - Analyze Remaining Useful Life (RUL) predictions.
    - Schedule maintenance operations.
    - Monitor equipment degradation through simulated real-time monitoring.
  - Needs:
    - Accurate Remaining Useful Life predictions.
    - Real-time monitoring dashboard.
    - Explainable prediction results.
    - Maintenance recommendations.
    - Automatic real-time equipment monitoring.
    - Continuous Remaining Useful Life updates.
  - Priority:
    - High.

- **Plant Managers**
  - Role:
    - Plant managers oversee operational performance and maintenance planning to ensure efficient production and minimal downtime.
  - Responsibilities:
    - Monitor operational performance.
    - Approve maintenance schedules.
    - Evaluate maintenance costs.
    - Improve operational efficiency.
  - Needs:
    - Equipment health reports.
    - Downtime statistics.
    - Performance metrics.
    - Maintenance planning support.
    - Operational dashboards for decision support.
  - Priority:
    - High.

- **System Administrators**
  - Role:
    - System administrators are responsible for deploying, maintaining, and monitoring the predictive maintenance platform.
  - Responsibilities:
    - Deploy backend services.
    - Monitor API availability.
    - Ensure system reliability.
    - Monitor application logs.
  - Needs:
    - Stable deployment.
    - Secure APIs.
    - Reliable performance.
    - Error monitoring.
  - Priority:
    - Medium.

- **Machine Learning Engineers**
  - Role:
    - Machine learning engineers develop and improve machine learning models for Remaining Useful Life prediction.
  - Responsibilities:
    - Analyze data.
    - Engineer features.
    - Train machine learning models.
    - Evaluate model performance.
    - Implement Explainable AI.
    - Monitor model performance during simulated real-time prediction.
  - Needs:
    - High-quality datasets.
    - Feature importance analysis.
    - Performance metrics.
    - Model explainability.
  - Priority:
    - High.

- **Project Team**
  - Role:
    - Responsible for designing, implementing, testing, documenting, and integrating all project components.
  - Responsibilities:
    - Implement system modules.
    - Integrate backend and frontend.
    - Conduct testing.
    - Prepare documentation.
  - Needs:
    - Clear requirements.
    - Version control.
    - Task coordination.
  - Priority:
    - High.
- **DEPI Evaluators**
  - Role:
    - Evaluate the project based on technical quality, documentation, innovation, teamwork, and presentation.
  - Needs:
    - Professional documentation.
    - Reliable implementation.
    - Explainable AI integration.
    - Demonstration of business value.
  - Priority:
    - High.

User Stories:

- **US-1: Maintenance Engineer**
  - As a Maintenance Engineer, I want to predict the Remaining Useful Life (RUL) of equipment so that I can schedule maintenance before failures occur and minimize unexpected downtime.
- **US-2: Maintenance Engineer**
  - As a Maintenance Engineer, I want to view historical equipment data and degradation trends so that I can analyze equipment behavior and support maintenance decisions.
- **US-3: Maintenance Engineer**
  - As a Maintenance Engineer, I want the system to automatically stream simulated sensor data so that I can continuously monitor equipment health and Remaining Useful Life (RUL) without manual intervention.
- **US-4: Maintenance Engineer**
  - As a Maintenance Engineer, I want the dashboard to update sensor readings, equipment health status, and RUL predictions automatically during the simulation so that I can monitor equipment degradation without manual intervention.
- **US-5: Maintenance Engineer**
  - As a Maintenance Engineer, I want to select a dataset scenario and engine so that the system automatically loads the corresponding sensor data for prediction and monitoring.
- **US-6: Plant Manager**
  - As a Plant Manager, I want to understand the reasons behind model predictions so that I can trust the system and make informed maintenance decisions.
- **US-7: Plant Manager**
  - As a Plant Manager, I want to generate maintenance and performance reports so that I can evaluate equipment reliability and maintenance effectiveness.

- **US-8: Plant Manager**
    - As a Plant Manager, I want to view feature importance and SHAP explanations so that I can understand why the model generated a specific Remaining Useful Life prediction.

Use Cases:

- **UC-1: Dataset Selection**
    - Primary Actor
        - Maintenance Engineer.
    - Preconditions
        - The NASA CMAPSS dataset is available.
        - The system is initialized.
    - Main Flow
        - User selects FD001–FD004.
        - User selects Engine ID.
        - System validates selection.
        - System validates the selected dataset and engine.
        - Selected data is prepared for simulation and prediction.
    - Output
        - Selected dataset.
- **UC-2: Automatic Real-Time Monitoring**
    - Primary Actor
        - Maintenance Engineer.
    - Preconditions
        - Dataset selected.
        - A trained prediction model is available.
        - Simulation started.
    - Main Flow
        - System automatically streams sensor readings.
        - Dashboard updates every cycle.
        - Backend predicts RUL continuously.
        - Equipment health status is updated.
        - Visualization refreshes automatically.
    - Output
        - Continuous equipment monitoring.
- **UC-3: Predict Remaining Useful Life (RUL)**
    - Primary Actor
        - Maintenance Engineer.
    - Preconditions
        - A trained machine learning model is available.
        - A NASA CMAPSS dataset scenario has been selected.
        - The backend API is running and accessible.

- Main Flow
  - The system automatically loads the corresponding sensor data.
  - The backend preprocesses incoming data.
  - The machine learning model predicts the Remaining Useful Life (RUL).
  - The dashboard automatically displays updated predictions.
- Output
  - Predicted Remaining Useful Life (RUL).

- **UC-4: Dashboard Visualization**
  - Primary Actor
    - Maintenance Engineer.
  - Preconditions
    - A prediction has been generated.
    - The dashboard is connected to the backend service.
  - Main Flow
    - Dashboard receives prediction.
    - Dashboard updates charts.
    - Dashboard updates gauges.
    - Dashboard updates health indicator.
    - Dashboard displays SHAP visualization.
  - Output
    - Updated dashboard.

- **UC-5: View Historical Sensor Data**
  - Primary Actor
    - Maintenance Engineer.
  - Preconditions
    - Historical sensor data is available.
  - Main Flow
    - The user selects dataset scenario.
    - The user selects engine.
    - The user selects operational cycle range.
    - The system retrieves historical sensor measurements.
    - The dashboard displays sensor trends and degradation visualizations.
  - Output
    - Historical sensor readings and trend analysis.

- **UC-6: Explain Prediction**
  - Primary Actor
    - Maintenance Engineer.
  - Preconditions

        - A prediction has been generated.
    - Main Flow
        - The user requests an explanation for the prediction.
        - The SHAP framework computes feature contributions.
        - The dashboard visualizes feature importance and SHAP values.
        - The dashboard highlights the most influential sensors affecting the prediction.
        - The user reviews and interprets the factors influencing the prediction.
    - Output
        - Feature importance analysis and SHAP explanations.

- ## UC-7: Review Maintenance Recommendation
    - Primary Actor
        - Maintenance Engineer.
    - Preconditions
        - A Remaining Useful Life (RUL) prediction has been generated.
        - Equipment health status has been evaluated.
    - Main Flow
        - User reviews RUL prediction.
        - User reviews equipment health status.
        - System provides maintenance recommendation.
        - User reviews the recommendation and decides whether maintenance actions should be initiated.
    - Output
        - Maintenance recommendation.

- ## UC-8: Generate Reports
    - Primary Actor
        - Plant Manager.
    - Preconditions
        - Prediction and maintenance data are available.
    - Main Flow
        - The user selects the report type.
        - The system generates the requested report.
        - The report includes prediction summaries, model performance metrics, explainability analysis, and maintenance recommendations.
    - Output
        - Maintenance and performance reports.

Functional Requirements:

- **Dataset & Simulation**
  - FR-01
    - The system shall allow users to select a NASA CMAPSS dataset scenario (FD001–FD004).
  - FR-02
    - The system shall allow users to select an engine from the selected dataset.
  - FR-03
    - The system shall automatically load and validate the corresponding sensor data before starting the simulation.
  - FR-04
    - The system shall simulate real-time sensor data streaming by sequentially replaying operational cycles from the selected NASA CMAPSS dataset.
- **Prediction Engine**
  - FR-05
    - The system shall preprocess incoming sensor data using the same preprocessing pipeline applied during model training.
  - FR-06
    - The system shall predict the Remaining Useful Life (RUL) using the trained machine learning model.
  - FR-07
    - The system shall continuously update the Remaining Useful Life (RUL) prediction after each simulated operational cycle.
  - FR-08
    - The system shall classify equipment health status (Healthy, Warning, or Critical) based on the predicted Remaining Useful Life (RUL).
- **Explainable AI**
  - FR-09
    - The system shall generate SHAP-based explanations for each prediction to identify the contribution of individual sensor features.
  - FR-10
    - The dashboard shall display SHAP values, feature importance rankings, and explanation visualizations for every prediction.

- **Dashboard**
  - FR-11
    - The system shall provide an interactive Streamlit dashboard for monitoring equipment health and prediction results.
  - FR-12
    - The dashboard shall automatically update sensor readings during simulation.
  - FR-13
    - The dashboard shall display (RUL, Equipment Health Status, Sensor Trends, SHAP Explanations, & Current Simulation Cycle).
- **Historical Analysis**
  - FR-14
    - The system shall allow users to view historical sensor data.
  - FR-15
    - The system shall visualize equipment degradation trends across operational cycles using interactive charts.
- **Backend**
  - FR-16
    - The backend shall expose RESTful API endpoints for prediction requests and dashboard communication.
  - FR-17
    - The backend shall send prediction results, equipment health status, and SHAP explanations to the dashboard in real time during simulation.
  - FR-18
    - The system shall load the trained machine learning model automatically when the backend service starts.
- **Reports**
  - FR-19
    - The system shall generate downloadable maintenance and performance reports.
  - FR-20
    - The generated reports shall include (Prediction Summary, Model Performance Metrics, SHAP Explainability Analysis, Equipment Health Status, & Maintenance Recommendation).

Non-Functional Requirements:

- **Performance**
  - NFR-01
    - The system shall generate Remaining Useful Life (RUL) predictions within one second after receiving valid sensor data under normal operating conditions.
  - NFR-02
    - The dashboard shall load completely within three seconds under normal operating conditions.
- **Availability**
  - NFR-03
    - The system shall remain available and responsive throughout each simulation session without unexpected interruptions.
- **Reliability**
  - NFR-04
    - The system shall produce consistent prediction results when processing identical input data and model configurations.
- **Security**
  - NFR-05
    - The system shall validate and sanitize all user inputs, rejecting invalid dataset selections, engine IDs, and malformed requests before processing.
- **Usability**
  - NFR-06
    - The system shall provide a simple and intuitive user interface.
  - NFR-07
    - Predictions, equipment health status, SHAP explanations, simulation progress, and visualizations shall be clearly displayed and easy to interpret.
- **Maintainability**
  - NFR-08
    - The codebase shall follow PEP8 coding standards.
  - NFR-09
    - The system shall follow a modular architecture to support maintainability and extensibility.

- **Scalability**
  - NFR-10
    - The system architecture shall support adding new datasets and machine learning models with minimal code modifications.
- **Portability**
  - NFR-11
    - The system shall be executable on Windows, Linux, and macOS environments supporting Python.
- **Compatibility**
  - NFR-12
    - The system shall be compatible with modern web browsers including Google Chrome, Microsoft Edge, and Mozilla Firefox.
- **Explainability**
  - NFR-13
    - The system shall provide clear and consistent SHAP-based explanations for every prediction.
- **Documentation**
  - NFR-14
    - The source code shall include clear comments, API documentation, and developer documentation to facilitate future maintenance and extension.
- **Reproducibility**
  - NFR-15
    - The system shall produce reproducible prediction results when executed using the same trained model, preprocessing pipeline, and input data.